(or any polygon's perimeter) is a valid Jordan curve. For graphs' edges we'll use sections of Jordan curves, so the only restriction will actually be that it must not self-intersect in any point.

Drawing a graph on a plane can be thought of as placing vertices on a 2D Euclidean space. Informally, we can imagine each vertex as a point in $\mathbb{R}^2$ (the set of all pairs of real numbers), and each edge as an arc (or a polyline) between two vertices.

More formally, we can define a planar embedding as an isomorphism (a 1:1 mapping) between an abstract graph G and a plane graph G'.

A plane graph, in turn, is defined as a pair of finite sets (V, E), as always denoted as vertices and edges respectively, such that:

1. V is a subset of $\mathbb{R}^2$.
2. Every edge $e \in E$ is a section of a *Jordan curve* passing through two vertices.
3. No two edges have the same pair of endpoints.
4. No edge intersects a vertex (other than its endpoints) nor any other edge.

We still owe you a definition: a Jordan curve is as a plane, simple and closed curve, a non-self-intersecting continuous loop in the plane. See figure 15.4 for a few examples.

A planar graph is thus defined as an abstract graph G for which it exists a planar embedding.